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12 December 1984
SUPERSEDING
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1 March 1979

MILITARY SPECIFICATION

WATER PURIFICATION UNIT, REVERSE OSMOSIS, 600 GPH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a military design reverse osmosis water purification unit having a nominal capacity of 600 GPH on fresh water.

1.2 Classification. The water purification units shall be of the following types, as specified (see 6.2.1):

Type I - Trailer mounted.
Type II - Skidmounted: two fork pocket.
Type III - Skid mounted: four fork pocket.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4610

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment.)
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3659	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-W-6858	- Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-A-8421	- Air Transportability Requirements, General Specification for.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-19834	- Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear Actuator Screw.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies: Preservation and Packaging of.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-810[1/]	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS49006	- Plugs, Pipe, Magnetic-Headless; Aluminum, Iron or Steel.

DRAWINGS

ME

TA13222E5235	- Water Purification Unit, Reverse Osmosis, Trailer-Mounted, 600 GPH.
TA13225E8940	- Water Purification Unit, Reverse Osmosis, Skid Mounted, Four Fork Pocket, 600 GPH.
TA13226E0121	- Water Purification Unit, Reverse Osmosis, Skid Mounted, Two Fork Pocket, 600 GPH.

PUBLICATIONS

TM5-4610-215-10	- Water Purification Unit, Reverse Osmosis, 600 GPH, Trailer Mounted, Flat Bed Cargo, 5 Ton, 4 Wheel Tandem (Model ROWPU 600-1)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

1/ The "C" revision of MIL-STD-810 (dated 10 March 1975) shall be used.

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CODE OF FEDERAL REGULATIONS (CFR)

TITLE 21- Federal Food, Drug & Cosmetic, Food Additive.

(Application for copies of the above publication is available from the Superintendant of Document, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY (AWS)

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.9 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Welding Rods and Bare Electrodes.
A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1141 - Substitute Ocean Water.
D3951 - Standard Practices for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Useage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 East Flair Drive, El Monte, CA 91731.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The water purification units shall be in accordance with top assembly, TA13222E5235 for type I, and top assembly, TA13226E0121 for type II, and top assembly, TA13225E8940 for type III and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the water purification unit. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article . When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings (see 6.13). Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The material shall have no adverse effect on the health of personnel when used for its intended purpose. The lines and all surfaces that contact potable water shall conform to the Federal Food, Drug and Cosmetic, Food Additives, Title 21, Code of Federal Regulations for contact with food. The compounds listed in section 177.1020, 177.1030, 177.1040, 177.1050 or 177.1480 of the Federal Food, Drug and Cosmetic, Indirect Food Additives (Chapter 1, Title 21, Code of Federal Regulations), shall not be acceptable materials. Questions pertinent to this effort shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 6.4).

3.4 Operation. The water purification unit shall operate at design capacity without leaks, malfunction, or permanent deformation. The water purification unit, when operated in accordance with the applicable technical manual, shall have a water production capacity of 540-660 gph on fresh water containing less than 500 ppm of total dissolved solids at a water temperature of 77° F. Production rate for sea water shall be a minimum of 25 percent of the fresh water rate when tested as pecified in 4.6.2.9.

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3.4.1 Total dissolved solids rejection. The water purification unit shall produce product water containing a maximum of 1500 ppm total dissolved solids when tested as specified in 4.6.2.3. and 4.6.2.9.2.

3.5 Environmental.

3.5.1 Operating temperature. The water purification unit shall perform as specified in any ambient temperature from plus 110° F to minus 25° F when tested as specified in 4.6.2.5. Shelter is to be used when the temperature is below 32° F to provide the required protection for operation down to minus 25° F.

3.5.2 Storage temperature. The water purification unit shall not be damaged by storage at ambient temperatures from plus 160° F to minus 30° F when tested as specified in 4.6.2.5. The reverse osmosis elements and calcium hypochlorite shall be removed from the unit during storage.

3.6 Safety. The water purification unit shall not present any safety hazard to operating and maintenance personnel and shall include, but not be limited to, the following general safeguarding provisions:

a. All reciprocating, rotating, or moving parts of the equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard.
b. All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations.
c. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.
d. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.
e. Access to the chemical feeder shall be as to not cause splashing of chemicals.
f. Each container of chemicals (citric acid, polyelectrolyte, surfactant, sodium hexametaphosphate) shall be labeled with the generic name and the following note:

"CAUTION: Protect the eyes and skin from direct contact with chemicals. Be especially careful when opening containers to avoid inhaling chemical powder. Wash hands thoroughly in soap and water after handling chemicals."

Labels shall be on identification plates conforming to MIL-P-19834, type 1, style 2.

3.7 Human factors. The water purification unit shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472 where practicable. The water purification unit shall be operable and maintainable by the 5th to 95th percentile.

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3.7.1 Noise level. The noise level of the water purification unit (excluding horns and warning alarms shall be in accordance with MIL-STD-1474, category D, for steady-state noise when tested in accordance with 4.6.2.8. Hearing protectors shall not be supplied with the water purification unit. If the noise levels at the operator station, with the generator running at normal capacity, exceeds 85 dB(A), the provisions of MIL-STD-1474 shall be followed. Hazard signs, if required, shall state "CAUTION, HEARING PROTECTION REQUIRED WITHIN _____feet." Plates shall conform to MIL-P-514, type 1, style 1, composition C of type I, grade A, class I material (see 6.5).

3.8 Reliability. The specified mean-time-between-failure (MTBF) of the water purification unit and ancillary equipment (not including generator) shall be 500 hours when tested as specified in 4.6.2.9. During reliability testing, the trailer shall exhibit no failures.

3.9 Maintainability. The water purification unit shall have a maintenance ratio of not more than 0.03. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing. Generator shall not be included in maintenance ratio. (see 4.6.2.10)

3.10 Electromagnetic interference. The electromagnetic interference emission characteristics of the water purification unit shall conform to MIL-STD-461, CLASS C1, Tactical and Special Purpose Vehicles and Engine-Driven Equipment.

3.11 Trailer electrical system. All lights and electrical circuits shall be capable of operation throughout all vehicle operating conditions.

3.12 Trailer brake system. The trailer shall be equipped with an automatically engaging service brake which allows the brakes to engage when the intervehicular hoses are disconnected. The automatically engaging service brakes shall maintain application of brakes for not less than 15 minutes when intervehicular hoses are disconnected.

3.13 Leveling jacks. The manually operated jacks shall have the capacity to raise and support the trailer and shall provide adjustment to permit placing the jack bases to the ground.

3.14 Lubrication.

3.14.1 Lubricants. The water purification unit shall operate as specified herein when serviced with military lubricants (see 6.6).

3.14.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

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3.14.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.14.4 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2 inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drainplug, size conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accomplished when the water purification unit is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet.

3.14.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments, cameras, electronic gear and mechanical control systems; for use on gears, actuator screws and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.
b. MIL-G-10924 - For lubrication and surface corrosion protection of automotive and artillery equipment operated over the temperature range of minus 65° F to plus 225° F. Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable. Not for use on machinery which comes in contact with foods or potable water.

The water purification unit shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.14.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), and upon completion of all testing, all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F to plus 32° F. A tag shall be attached in a conspicuous place to indicate the temperature range, and the lubricant grade used.

3.15 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.15.1, with 1 percent

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copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.15.1 Composition. Composition I shall be used unless composition II is required in order to comply with local air pollution regulations governing the application of the varnish. When composition II is used, the contractor shallprovide evidence to the Government that the use of composition II is required and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.16 Identification marking. The water purification unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the water purification unit on plates conforming to MIL-P-514, type III, composition C, type I or II, class 1 material. Information included shall be as specified on the appropriate drawing. The plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.17 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material shall be provided. In addition, the plate shall show the silhouette of the water purification unit in transport position indicating the center of gravity and the location and capacity of the slinging and tiedown provisions. The plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.18 Instruction plates. Each water purification unit shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the water purification unit. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plates will be both visible and legible.

3.19 Treatment and painting. Treatment, painting, and color shall be in accordance with MIL-T-704 or the applicable drawing.

3.19.1 Dissimilar metals. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

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3.20 Transportability.

3.20.1 Air. The water purification unit shall conform to MIL-A-8421 for both air transportability requirements and airdrop requirements.

3.20.2 Rail. The water purification unit shall have a dimensional profile in accordance with the International Loading Gage. In addition, the trailer mounted water purification unit shall withstand rail impact without damage or permanent deformation to any part or component when tested as specified in 4.6.2.12.

3.20.3 Highway. The water purification unit shall withstand shock and vibration without damage or permanent deformation to any part or component when tested as specified in 4.6.2.12.

3.21 Government-furnished property. The following property in the quantities indicated will be furnished by the Government (see 6.15):

Item No.	Description	Identification	Quantity for each water purification unit
1	Water Purification Unit, Reverse Osmosis, 600 GPH, Trailer-Mounted, Flat Bed Cargo, 5 Ton, 4 Wheel Tandem (Model ROWPU 600-1)	TM5-4610-215-10	1

3.22 Vehicle weight classification. The contractor shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number to the water purification unit as requested (see 6.7).

3.23. Workmanship. All parts, components, and assemblies of the water purification unit including castings, forgings, molded parts, stamping, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered. The unit shall be free from such defects as misaligned components, incomplete welds, rust, extraneous material, cracks, leaks, and other defects that could impair the operation and serviceability of the unit.

3.23.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly.

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External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.23.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.23.3 Riveted connections. Rivets shall fill the holes completely. All rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.23.4 Welders, weldments, and welding.

3.23.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall develop a certification of welder qualification for the welder or welders (see 6.2.2).

3.23.4.2 Weld filler metals. Aluminum filler metal shall be in accordance with AWS A5.10, type ER5356 or ER5556, except that filler metal ER4043 may be used in highly restricted locations if conditions cannot be eliminated by adjustments of procedures or fixtures; or when joining base metal type 6061 to other types of aluminum. Steel filler metal shall be in accordance with AWS A5.1, Type E6010 or E6011 for carbon steel; Type E7018 for high-strength, low-alloy steel; type E6018 for high strength, low-alloy steel to carbon steel; and AWS A5.9, Type ER308L for corrosion-resisting steel.

3.23.4.3 Aluminum and steel weldments. The surfaces and edges of parts to be welded including each layer of weld materials when multiple layers are required shall be free from oxide scale, paint, lubricant, and other foreign matter. Welds shall transmit stress to the heat-affected zone without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. All welded parts shall be free from cracks, porosities and other imperfections that may reduce the effectiveness of the part. Welds involving corrosion-resistant steel shall be passivated and annealed as specified on the applicable drawings.

3.23.4.3.1 Aluminum welding. Aluminum welding, except spot and seam welding, shall be done by the metal-inert-gas (MIG) method or other methods that will provide equivalent mechanical properties of the filler metal to base metal combinations as specified in 3.22.4.2. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds without metallic or nonmetallic inclusions. Finished aluminum welds may be ground, filed, chipped, or wire-brushed, but shall not be hammered. Preheating for welding is permissible provided the temperature does not exceed 300° F for a total time not to exceed 30 minutes. No process requiring the use of welding flux shall be used. Work shall be positioned and rigged for flat welding whenever practicable.

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3.23.4.3.2 Steel welding. Steel welding shall be done by the shielded-electric-arc method or metal-inert-gas (MIG) process in accordance with qualification procedures prescribed in 3.22.4.1, or any other method providing equivalent mechanical properties of the filler metal to base metal combinations as specified in 3.23.4.2. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed. Work shall be positioned for flat welding whenever practicable. Complete and uniform penetration and fusion of the metals shall be obtained on all welds.

3.23.4.4 Weldment specimens. The contractor shall submit steel and aluminum specimens representing each typical weld joint utilized in fabrication of the water purification unit. These specimens shall be furnished at the same time as the first article water purification unit. Weldment specimens shall be submitted by each welder performing weldments on the water purification unit with each welding equipment used. Acceptable specimens shall represent the minimum acceptable weld quality and shall be retained as an inspection standard for evaluation of production weldments.

3.23.4.5 Welding procedure. Prior to any welding, the contractor shall prepare and submit to the contracting officer for approval, a written welding procedure defining process and control. This procedure shall include, but not be limited to, the factors listed in table I. Welding procedure changes affecting any of the factors in table I shall be submitted to the contracting officer for approval prior to adoption of the change.

TABLE I. Factors to be included in joint welding procedure.

Item No.	Factor
1.	Welding process and techniques including shielding medium.
2.	Welding process operation.
3.	Type of current, ac or dc; and polarity of dc, if used.
4.	Shielding gas composition.
5.	Joint design.
6.	Method of joint preparation.
7.	Preweld and interpass cleaning.
8.	Preheat or ambient temperature range.
9.	Composition of filler metal.
10.	Filler metal size for all passes.
11.	Number and sequence of passes.
12.	Welding position.
13.	Repair method to be used in removing defects.
14.	Method of inspection used to insure proper repair.
15.	Method of qualification of welders.
16.	Method of determining capability of automatic equipment and equipment operators.
17.	Workmanship specimens.
18.	Method of inspection of finished welds.

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3.23.4.6 Rewelding defects. Portions of welded joints that have been rejected on inspection because of defects specified in 3.23.4.3 or not conforming to the weldment specimens specified in 3.23.4.4 may be repaired only by rewelding. The defective weld shall be removed by chipping or machining. With the exception of steel, flame cutting shall not be used. Before rewelding, the joint shall be inspected to assure that all of the defective weld has been removed and that the joint is accessible so that the welding operator can obtain full penetration and fusion through the joint. Repairs in the form of patching aluminum shall not be performed.

3.23.4.7 Spot and seam welding. Spot and seam welding shall be in accordance with MIL-W-6858.

3.23.4.8 Brazing. Brazing of steels, copper alloys, and nickel alloys shall conform to MIL-B-7883.

3.23.5 Diagrams, charts, and plates. Diagrams, charts, and plates attached to components with adhesive shall be applied to give a surface free of wrinkles, bubbles, or other defects that may cause the diagrams, charts, and plates to become loose or damaged.

3.23.6 Cleaning. The water purification unit shall be throughly cleaned of all cutting oil, residue of tests, grease, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that will not leave a residue or otherwise make the water purification unit unsuitable for its intended use. Except where moisture is required for preservation during shipment, the unit shall be completely drained before preparation for packaging.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Water purification units which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations and tests in section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

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4.1.3 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.4 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.8 and 6.9.1).

4.1.5 End item final inspection requirement (EIFIR). Each water purification unit shall be inspected in accordance with EIFIR-EF-13222E5235. One copy of the completed EIFIR shall be shipped with each inspected water purification unit (see 6.9.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 First article exmination. The first article shall be examined as specified in 4.5.1 in the order shown. Presence of one or more defects shall be cause for rejection or shall be cause for performing the disassembly inspection specified in 4.1.3.

4.3.2 First Article tests. Upon successful completion of the examination specified in 4.3.1, the first article shall be subjected to the tests marked "X" in column 1 of table III. Failure of any test shall be cause for rejection or shall be cause for performing the disassembly inspection specified in 4.1.3.

4.4 Inspection failure. Failure of a first article water purification unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3 shall be cause for rejection of the first article water purification unit and shall be cause for refusal by the Government to continue acceptance of production water purification units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on water purification units previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be

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considered prima facie evidence that all water purification units accepted prior to the completion of first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each water purification unit shall be examined for the characteristics marked "X" in column 2 of table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Upon successful completion of the examination 4.5.1, each water purification unit shall be subjected to the tests marked "X" in column 2 of table III. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table II.

TABLE II. Examination schedule.

First Article	Qual. Conf.	Characteristics	Requirement Paragraph
1	2	3	4
X	X	101. Any part or component not in accordance withthe QAP requirements as shown on the drawings.	Applicable dwgs.
X	X	102. Any dimension, other than those identified by the QAP requirements shown on the drawings, not as specified.	Applicable dwgs.
X	X	103. Assembly not complete or not correct.	Applicable dwgs.
X	X	104. Parts or components missing.	Applicable dwgs.
X	X	105. Parts or components not as specified.	Applicable dwgs.
X	X	106. Material not as specified.	Applicable dwgs.
X	X	107. Treatment and painting not as specified.	3.19 and applicable dwgs.
X	X	108. Lubrication not as specified.	3.14
X	X	109. Lubricants not as specified.	3.14.1
X	X	110. Lubrication fittings not as specified.	3.14.2
X	X	111. Pressure-release device not as specified.	3.14.3

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TABLE II. Examination schedule. Cont'd

First Article	Qual. Conf.	Characteristics	Requirement Paragraph
1	2	3	4
X	X	112. Filling, draining, and checking provisions not as specified.	3.14.4
X	X	113. Grease lubrication not as specified.	3.14.5
X	X	114. Oil lubrication not as specified.	3.14.6
X	X	115. Workmanship not as specified.	3.23, 3.23.1, 3.23.5
X	X	116. Water purification unit not completely drained and cleaned before preparation for packaging.	3.23, 3.23.6
X	X	117. Identification not as specified	3.16
X	X	118. Shipping data plates missing, not attached or illegible, as specified.	3.17
X	X	119. Instruction plates missing, not attached or illegible, as specified.	3.18
X	X	120. Safety not as specified.	3.6
X	X	121. Chemical containers not labeled as specified.	3.6
X	X	122. Reflectors on trailer not as specified.	Applicable dwgs.
X	X	123. Air brakes not as specified.	Applicable dwgs.
X	X	124. Frame not as specified.	Applicable dwgs.
X	X	125. Electrical installation not as specified.	Applicable dwgs.
X	X	126. Lunette not as specified.	Applicable dwgs.
X	X	127. Chain assembly not as specified.	Applicable dwgs.
X	X	128. Bumper stops not as specified.	Applicable dwgs.
X	X	129. Spare tire and wheel assembly not as specified.	Applicable dwgs.
X	X	130. Suspension assembly not as specified.	Applicable dwgs.
X	X	131. Swivel assembly not as specified.	Applicable dwgs.
X	X	132. Jack assembly not as specified.	Applicable dwgs.

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TABLE II. Examination schedule. Cont'd

First Article	Qual. Conf.	Characteristics	Requirement Paragraph
1	2	3	4
X	X	133. Generator mounting holes not as specified.	Applicable dwgs.
X	X	134. Protection against fungus and moisture resistance not as specified.	3.15
X	X	135. Vehicle weight classification is missing.	3.21

4.6.2 Tests.

4.6.2.1 Test schedule. Tests shall be accordance with table III.

TABLE III. Test schedule.

First Article	Qual. Conf.	Test	Requirement	Test Paragraph
1	2	3	4	5
X	-	Operational (40 Hours).	3.4	4.6.2.3
-	X	Operational (6 Hours).	3.4	4.6.2.4
X	-	Environmental.	3.5	4.6.2.5
X	-	Safety.	3.6	4.6.2.6
X	-	Human factors.	3.7	4.6.2.7
X	-	Noise level.	3.7.1	4.6.2.8
X	-	Reliability (MTBF).	3.8	4.6.2.9
X	-	Maintainability.	3.9	4.6.2.10
X	-	Electromagnetic interference.	3.10	4.6.2.11
X	-	Transportability.	3.20	4.6.2.12
X	X	Trailer electrical system.	3.11	4.6.2.13
X	X	Trailer brake system.	3.12	4.6.2.14
X	X	Leveling jacks.	3.13	4.6.2.15

4.6.2.2 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the environmental conditions existing at the place of test. The water purification unit shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. Prior to start of test, components of the water purification unit requiring lubrication shall be lubricated with oils and greases specified in 3.14. Oils shall be those designated for use in the ambient temperature at the place of test.

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4.6.2.3 Operational test (40 hours). The first article water purification unit shall be operated with substitute ocean water (chlorine free) for not less than 40 hours as specified in 4.6.2.3.1 through 4.6.2.3.2. At least 5 back-washing cycles shall be performed during the 40 hour period. Upon completion of the test, the unit shall be flushed with fresh chlorine free water for a period of 10 minutes. Fill the chemical utility pails with fresh water. Set all control knobs on the chemical feed pump to 10. Operate the unit in a normal manner during the 10 minute period. After flushing, open and leave open all vent and drain valves. Follow the instructions in Para. 2-30, preparation for movement, of TM 5-4610-215-10 for draining all components of the ROWPU.

4.6.2.3.1 Test equipment. Test equipment shall include but shall not be limited to the following:

Power source - 120/208-volt, 3-phase, 4-wire, 60-Hz.
Raw water pump - self priming, 30 gpm at 120 foot head.
Backwash pump - centrifugal, 120 gpm at 75 foot head.
Distribution pump - 30 gpm at 50 foot head.
Suction hose - NPSH couplings.
Calcium hypochlorite - Commercial, 65 percent available chloride, minimum.
Polyelectrolyte, cationic - Catfloc TI.
Test feedwater - ASTM D 1141, substitute ocean water, formula A, section 6 without heavy metals, shall be used. The salt shall be dissolved in dechlorinated tap water to provide a solution containing 19,000 ± 1000 mg/l, chloride ion. Sodium thiosulfate may be used for removal of chlorine. The permeate and concentrate streams may be blended for recirculation provided that the temperature of the feed does not exceed 90° F.

4.6.2.3.2 Test procedure. The water purification unit with ancillary equipment (see 6.14) shall be set up, operated, and maintained in accordance with the applicable technical manual.

4.6.2.3.3 Criteria for failure. Evidence of any one of the following conditions shall constitute failure of this test:

a. Leaks at any welded seam or any connection.
b. Light, or any switch fails to operate.
c. Valves fail to operate properly.
d. Mechanical or electrical failure of any component or part.
e. Failure of the unit to exhibit a total dissolved solids rejection as specified in 3.4.1.
f. Failure of unit to exhibit a production rate specified in 3.4.

4.6.2.4 Operational test (6 hours). The water purification unit shall be tested as specified in 4.6.2.2 and 4.6.2.3.1 through 4.6.2.3.3 for not less than 6 hours except that condition (e) of 4.6.2.3.3 shall not constitute criteria for failure of the 6 hour operational test. Upon completion of the test, the ROWPU shall be flushed with fresh chlorine free water for a period of 10 minutes. Fill

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the chemical utility pails with fresh water. Set all control knobs on the chemical feed pump to 10. Operate the unit in a normal manner during the 10-minute period. After flushing, open and leave open all vent and drain valves. Follow the instructions in Para 2-30, preparation for movement, of TM 5-4610-215-10 for draining all components of the ROWPU.

4.6.2.5 Environmental.

4.6.2.5.1 High temperature. Water purification units shall be tested in accordance with MIL-STD-810C, method 501, procedure I. The maximum test temperature for storage shall be 160° F, and for operating shall be 110° F. Operation shall be in accordance with the 6-hour operational test specified in 4.6.2.4, except that condition (e) of 4.6.2.3.3 shall be criteria for failure.

4.6.2.5.2 Low temperature. Test shall be conducted in accordance with MIL-STD-810C, method 502, procedure I. The minimum test temperature for storage shall be minus 30° F, and the minimum operating temperature shall be minus 25° F. Adequate shelter will be added to maintain the temperature at +35° ±2° F. Operation shall be in accordance with the 6-hour operational test specified in 4.6.2.4, except that condition (e) of 4.6.2.3.3 shall be criteria for failure.

4.6.2.6 Safety. Safety evaluation shall be accomplished throughout testing. Nonconformance to 3.6 shall constitute failure of this test.

4.6.2.7 Human factors. The water purification units shall be evaluated to determine the human factor characteristics. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.8 Noise level.

4.6.2.8.1 Operator's station. Measure the sound level at the operator's station in accordance with MIL-STD-1474 for steady-state noise while the unit is operating at rated capacity. Nonconformance to 3.7.1 shall constitute failure of this test. Also, if the noise is in excess of 85 dB(A), then determine the 85 dB(A) envelope around the system. The maximum distance in the 85dB(A) envelope shall be specified on the sign required by 3.7.1.

4.6.2.8.2 Exterior spectator position. Measure the exterior sound level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.7.1 shall constitute failure of this test.

4.6.2.9 Reliability. Using the MTBF specified in 3.8, the water purification unit shall be tested as specified in 4.6.2.9 and 4.6.2.9.2 for 1950 hours. If five or fewer failures occur, the unit shall be considered acceptable. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, degradation of performance capability of the equipment below designated levels, or serious personnel safety hazards. Any malfunction which the operator can remedy will not be considered a failure provided that the repair is authorized or prescribed as an operator function and can be accomplished within 60 minutes using only controls and

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authorized tools or spare parts incorporated in or carried with the equipment. Simultaneous related malfunctions are considered as one failure. Generator failures shall not be considered in accessing MTBF. The trailer shall exhibit no failures. If the trailer exhibits one or more failures, this constitutes failure of the test.

4.6.2.9.1 Test procedure. The water purification unit with ancillary equipment (see 6.14), shall be set up, operated, and maintained in accordance with the applicable technical manual supplied with the unit. Commence operation using the procedures outlined in the technical manual. Operate equipment continuously for periods of 20 hours per day at rated capacity, then shut down unit. Scheduled authorized operator, daily preventive maintenance and cleaning is allowed during the 4-hour non-operating period (not to be included in assessing MTBF). Any other actions will be used as failures in assessing MTBF.

4.6.2.9.2 Saltwater test. The water purification unit shall be tested as specified in 4.6.2.9 using a feedwater as specified in 4.6.2.3.1 for a minimum of 500 hours. The 500 hours shall be included as part of the 1950 hours.

4.6.2.10 Maintainability. Maintenance operations required during initial production testing shall be accomplished in accordance with 3.9. Non-conformance to 3.9 shall constitute failure of this test.

4.6.2.11 Electromagnetic interference. The water purification unit shall be tested to determine conformance to 3.10. Nonconformance to 3.10 shall constitute failure of this test.

4.6.2.12 Transportability. Water purification units shall be tested in accordance with MIL-STD-810c, method 516.2, procedure VI, and method 514.2, procedure XIII, part 2, except the unit shall be impacted once at the speeds of 4, 6, and 8 miles per hour ± 5 percent and once in the opposite direction at 8 miles per hour ± 5 percent (total of four impacts). Any visual damage to the water purification unit shall constitute failure of this test.

4.6.2.13 Trailer electrical system. Connect the intervehicular cable to a 24 volt power source. All lights shall be checked for proper operation. Nonconformance to 3.11 shall constitute failure of this test.

4.6.2.14 Trailer brake system. The intervehicular hoses shall be disconnected from an appropriate air source. The brakes shall engage and maintain application of brakes for not less than 15 minutes. Nonconformance to 3.12 shall constitute failure of this test.

4.6.2.15 Leveling jacks. Each jack shall be individually lowered and raised to support the trailer. Nonconformance to 3.13 shall constitute failure of this test.

4.7 Inspection of packaging.

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4.7.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

136.	Materials, methods or containers not as specified for level A. Each incorrect material, method or container shall constitute one defect.
137.	Test reverse osmosis elements or tubes not removed from the unit, the unit not drained and valves left open as specified for level A.
138.	Caps not installed on water inlets and outlets as specified for level A.
139.	Hardware items requiring a contact preservative not coated as specified for level A.
140.	Booster pump, RO pump and chemical feed pump not preserved as specified for level A.
141.	Ground rods not secured in the brackets as specified for level A.
142.	Trailer chassis not preserved as specified for level A.
143.	Raw water pumps, backwash pump and distribution pump not preserved as specified for level A.
144.	Canvas hoses not coiled and secured as specified, and hard rubber hoses not left straight and secured on the unit as specified for level A.
145.	Technical publications not preserved as specified for level A.
146.	Hand tools not preserved as specified for level A.
147.	Cotton duck case not preserved as specified for level A.
148.	Extension light cord not coiled or hanked and secured as specified for level A.
149.	Adapters, nipples, bushings, nozzle and strainer not wrapped with barrier material as specified for level A.
150.	Box closure for the comparator, conductivity meter and gate valves not method V as specified for level A.
151.	Flashlight not preserved as specified for level A.
152.	Elements not preserved in accordance with the referenced document specified for level A.
153.	Box of lamps not preserved as specified for level A.
154.	Cargo straps not preserved as specified for level A.
155.	Components not specifically preserved, not preserved as specified for components of similar design or construction as specified for level A.
156.	Consolidation of components and accessories in the chests not as specified for level A.

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157. Elements not consolidated together as specified for level A.
158. Calcium hypochlorite not consolidated as specified for level A.
159. Distribution pump for skid mounted units not boxed as specified for level A.
160. Other components not consolidated as specified for level A.
161. Preservation and packing not in accordance with the referenced document specified for commercial.
162. Trailer mounted units not fitted with a box type cover and the cover secured as specified for level A.
163. Skid mounted units not packed as specified for level A.
164. Marking missing, illegible, incorrect or incomplete for level A or commercial.
165. Filter tubes not installed as specified for level A.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2.1) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and appoval by the Government (see 6.3.1). The first article pack may be prepared utilizing either the first article model or a production water purification unit. When the first article model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model and production water purification units.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Test elements. Elements used for test purposes shall be removed from the unit; the unit drained and the drain valves left open. Install caps on water inlets and outlets.

5.2.1.2 Hardware items. Hardware items such as bolts, nuts, linkages, pins, grease fittings, and capscrews requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P1 or P19 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Booster pump, RO pump and chemical feed pump. The booster pump, RO pump and chemical feed pump shall be preserved in accordance with MIL-P-10603, level A. The RO pump housing and chemical feed pump housing shall be filled to the operating level with preservation lubricating oil conforming to MIL-L-21260, type 1, grade 10. The pumps shall be actuated to assure coating of all parts and surface within the housing.

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5.2.1.4 Ground rods. Ground rods shall be secured in the brackets provided on the unit.

5.2.1.5 Trailer chassis. The components of the trailer chassis shall be preserved in accordance with MIL-STD-281, level A (mobile).

5.2.1.6 Raw water pumps, backwash pump and distribution pump. These pumps shall be preserved in accordance with MIL-P-10603, level A.

5.2.1.7 Hose assemblies. Canvas hoses shall be coiled in a minimum size coil. The coils shall be secured with cotton tape or twine. Hard rubber hoses shall be left straight for consolidation on the unit.

5.2.1.8 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.9 Hand tools. Hand tools shall be preserved in accordance with PPP-P-40, level A.

5.2.1.10 Cotton duck case. The cotton duck case shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.11 Extension light. The cord shall be coiled or hanked to occupy minimum space and shall be secured with cotton tape or twine.

5.2.1.12 Adapters, nipples and bushings. Each unassembled adapter, nipple, and bushing shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 or wrapped with barrier material conforming to PPP-B-1055, class B-2 or MIL-B-121, type 1, grade A, class 2. The barrier material shall be secured with tape conforming to PPP-T-76.

5.2.1.13 Nozzle and strainer. The nozzle and strainer shall each be preserved in accordance with MIL-P-116, method IC-1 or IC-3 or wrapped with barrier material conforming to PPP-B-1055, class B-2, or MIL-B-121, type I, grade A, class 2. The barrier material shall be secured with tape conforming to PPP-T-76.

5.2.1.14 Comparator. The components of the comparator shall be placed in the case provided. The case containing the components shall be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, method V.

5.2.1.15 Conductivity meter. The conductivity meter shall be preserved in accordance with MIL-P-116, method 11b. The inner container shall be the vendor's carton or a box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125, style optional and the outer container shall be a box conforming to PPP-B-636, W6c, style optional. Box closure for the outer container shall be in accordance with the appendix to the box specification, method V.

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5.2.1.16 Flashlight. The flashlight shall be preserved in accordance with MIL-P-116, method IC-2 using the vendor's carton or a box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125, style optional.

5.2.1.17 Gate valves. Each gate valve shall be placed and secured in a close-fitting box conforming to PPP-B-636, W6c, style RSC. Box closure shall be in accordance with the appendix to the box specification, method V.

5.2.1.18 Elements. Elements shall be preserved in accordance with the requirements of the applicable drawing.

5.2.1.19 Box of lamps. The lamps shall be preserved in accordance with MIL-P-116, method IC-2, using the box provided as the container.

5.2.1.20 Cargo straps. Cargo straps not used for securing equipment in position shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.21 Filter tubes. One set of filter tubes shall be installed in the cartridge filter upon completion of testing. The other set (8 each) shall be placed in the storage box on the unit.

5.2.1.22 Other components. Other components not specifically mentioned herein requiring protection from corrosion or physical or mechanical damage shall be preserved as specified for components of similar design or construction.

5.2.1.23 Consolidation.

5.2.1.23.1 Chests 1 and 2. All components and accessories intended for the chests shall be placed in the applicable chest in accordance with the chest loading plan and the top closed and secured.

5.2.1.23.2 Elements. The preserved elements shall be consolidated together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.1.23.3 Calcium hypochlorite. The containers of calcium hypochlorite shall be consolidated together in a close-fitting box and the box closed and strapped as specified in 5.2.1.23.2.

5.2.1.23.4 Distribution pump. The distribution pump for type II or III skid mounted units shall be placed and secured in a close-fitting box and the box closed and strapped as specified in 5.2.1.23.2.

5.2.1.23.5 Other components. Other loose components that do not fit in the chests, shall be placed and secured on the unit as provided for, or if more convenient to accomplish the packing specified in 5.3 shall be consolidated together in a minimum number of close-fitting boxes as specified in 5.2.1.22.2. For trailer mounted units, the boxes shall be of a size and configuration best suited to secure on the unit in a manner that will not interfere with towing or lifting.

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5.2.2 Commercial. Each water purification unit, components, accessories and support equipment shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A or commercial as specified (see 6.2.1).

5.3.1 Level A.

5.3.1.1 Type I. Each complete type I (trailer mounted) water purification unit with all boxed or unboxed components, accessories and support equipment secured to the unit by the provided holddowns or with strapping conforming to QQ-S-781, type I or IV, size as applicable, finish A, shall be packed for mobile shipment. The basic unit shall be protected with a box type cover consisting of ends, sides and top constructed in accordance with PPP-B-601 for an overseas type, style I box. The box cover shall be the minimum size required to fit over and completely inclose the water purification structure and components extending above the trailer chassis; plywood shall be 3/8-inch thick. The sides of the cover shall be cut to follow the contour of the fender well area and shall not interfere with lifting or towing of the unit. A separate box cover constructed as specified herein for over the basic unit shall be provided over the distribution pump. The box covers shall be secured in place by the use of strapping, specified herein, wood members, nails, screws, nuts and bolts or any combination thereof that will provide securement of the covers.

5.3.1.2 Type II or III. Each complete type II or III (skid mounted) water purification unit, except the elements, calcium hypochlorite and distribution pump, (see 5.3.1.2.1), shall be packed in a crate conforming to MIL-C-104, type I, class 2, style a. Blocking, bracing and anchoring to prevent movement or damage of the contents shall be in accordance with the appendix to the crate specification.

5.3.1.2.1 Elements, calcium hypochlorite, and distribution pump. The preserved and consolidated elements, calcium hypochlorite and distribution pump shall not require additional packing but shall be shipped separate from but at the same time as the water purification unit for which intended.

5.3.2 Commercial. Each complete water purification unit with accessories and support equipment shall be packed in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 39518. In addition, weight and cube data shall be marked on the shipping container.

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6. NOTES

6.1 Intended use. The water purification unit is intended to supply potable water from any fresh, brackish, or sea water source.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of water purification unit required (see 1.2).
c. When a first article is required for inspection and approval, and the number of units required (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. Time frame required for submission of the first article pack (see 5.1).
f. Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, contract data requirements list (CDRL), the data requirements identified below shall be developed as specified by an approved data item description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverables data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirements	Applicable DID No.	Options
(a) 3.22.4.1	Certification, welder qualification	DI-H-24029	------

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of two or more units. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents' first articles.

6.3.1 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the water purification units in accordance with this specification.

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6.4 Medical advisor. The Surgeon General of the Army will act as advisor to make the necessary determination on health hazards. Inquiries should be forwarded to Headquarters, Department of the Army, ATTN: DASG-TSP-E, Washington, DC 20310.

6.5 Noise limits. When the noise level of the water purification unit exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include: (a) requirement for hearing protection; (b) the noise level of the water purification unit, (c) the distance from the water purification unit at which 85 dB(A) will always be met; and (d) what operational situation or system configuration will reduce the level to 85 dB(A).

6.6 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the water purification unit covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.7 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the contractor and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification number(s) for the water purification unit. The contractor should then be requested by the contracting officer to apply the weight classification number(s) to the water purification unit. The use of a Government-furnished sign kit conforming to MIL-S-40626 should be utilized where applicable (see 3.22).

6.8 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.9 Definitions.

6.9.1 Quality assurance provisions. A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.9.2 End item final inspection requirement (EIFIR). An EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance.

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6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.11 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each water purification unit.

6.12 Provisioning. The contracting officer should include provisioning requirements for such repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of water purification units.

6.13 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of the specification (see 3.3).

6.14 Ancillary equipment. The ancillary equipment shall consist of all items listed in the latest issue of SC 4610-97-CL-E16 (see 4.6.2.3.2 and 4.6.2.9.1).

6.15 Government-furnished property. The contracting officer should arrange to furnish the property in 3.22.

6.16 Decision flow chart. This decision flow chart is to be used only to determine those malfunctions which will be considered in the calculation of mean-time-between-failure (MTBF) and maintenance ratio (MR) (see 3.8).

Decision Flow Chart

Sequence	Question		Classification
1.	Does the incident concern RAM?	No	Non-RAM oriented No Score
	Yes		
2.	Does the incident concern the power source (commercial or generator)?	Yes	No Score
	No		
3.	Was incident detected during initial inspection?	Yes	No Score
	No		
4.	Did incident result from test item abuse, unrealistic operating conditions, accident, improper maintenance or operating procedures?	Yes	No Score
	No		

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Decision Flow Chart

Sequence	Question		Classification
5.	Was the incident a kit installation authorized modification, engineer evaluation, or Manual/Maintenance evaluation?	Yes	No Score
		No	
6.	Was the incident detected during an inspection which required only scheduled crew maintenance, if any, for correction?	Yes	No Score
		No	
7.	Was the incident a scheduled replacement/service?	Yes	Scheduled Maintenance No Score
		No	
8.	Was the incident caused by another incident?	Yes	Unscheduled Maintenance No Score
		No	
9.	Was the incident an incipient malfunction detected during scheduled maintenance and corrected at that level?	Yes	Unscheduled Maintenance No Score
		No	
10.	Was the incident an incipient malfunction detected during operation for which correction can be deferred to the next scheduled maintenance, and corrected at that level?	Yes	Unscheduled Maintenance No Score
		No	
11.	Was the incident, or could the incident have been corrected by operator/crew within 60 minutes using OEM (On Equipment Maintenance) tools and spare parts?	Yes	Unscheduled Maintenance No Score
		No	

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Decision Flow Chart

Sequence	Question		Classification
12.	Is the incident an actual malfunction for which maintenance can be deferred to the next scheduled maintenance and corrected at the level or deferred to the end of test? No	Yes	Unscheduled Maintenance No Score
13.	Could a ten hour work shift be completed without any degradation to operational mode mission essential functions, or critical or catastrophic hazard to personnel or equipment? No Failure Unscheduled Maintenance	Yes	Unscheduled Maintenance No Score

Custodian:
Army - ME
Air Force - 99

Review activities:
Army - MD, TE
Navy - YD, MC
Air Force - 84
DLA - CS

Preparing activity:
Army - ME

Project 4610-0097

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-52960A	Water Purification Unit, Reverse Osmosis 600 GPH

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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